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Green Customer Insights Platform · Lloyds · Megathon X: Singularity · Problem Statement 1

1. PROBLEM STATEMENT AND OUR UNDERSTANDING

The ask. Build an AI platform that analyses customer transactions, products and behaviour to (i) identify green financial opportunities, (ii) estimate their sustainability impact, (iii) give customers personalised insights, (iv) give business and relationship teams actionable insights, (v) encourage paperless journeys and (vi) **explain why** every opportunity was flagged. Four opportunities are required: **EV Finance, Green Home Loan, Solar Finance, Paperless Banking**; Green Investments and Green Loans are optional.

Our understanding.

- **One job per component.** ML models predict propensity; a rules-based carbon engine computes impact from documented factors; the LLM only explains structured outputs; eligibility is set by product rules. *A signal or propensity never establishes eligibility.*
- **We prepare our own data.** No real banking data: we generate synthetic customers, transactions and products with plausible, documented label-generation rules, customer-level train/validation/test splits and no leakage.
- **Trust is the product.** Each recommendation needs evidence, confidence, reason codes and a data timestamp (FR7), and must never suggest products the data rules out (e.g. EV finance for an EV owner with no replacement need).

2. IDEA / ABSTRACT

The platform turns transactions into green financial opportunities through a strictly decoupled **Neuro-Symbolic** design: **ML predicts, a rules engine calculates, and the LLM merely translates** validated, structured data into proactive advice, which removes hallucination risk from every number and eligibility decision a customer or banker sees.

LLM strategy: isolated from calculations

- **Schema-bound output:** FastAPI + Pydantic force a fixed JSON structure (headline, reasoning, impact, call-to-action).
- **Zero-temperature grounding:** temperature 0.0; prompts forbid inventing numbers, percentages or concepts outside the input JSON.
- **Proactive framing:** e.g. “Your monthly fuel spend is crossing £280. Switching to an EV could reduce your annual footprint by 1.2 tonnes.”
- **Output guardrail:** a backend validator checks every number the LLM returns against the engine's values; any mismatch triggers a rule-based text-template fallback.

Deterministic carbon engine (GHG Protocol / PCAF, no LLM)

- **Documented factors:** petrol 2.31 kg CO_2e/litre; grid electricity 0.207 kg CO_2e/kWh; paper statement 0.011 kg CO_2e.
- **Auditable:** results rounded to exact kg CO_2e and tagged with method (e.g. Scope 1/2 avoided emissions), with assumptions, baseline and time horizon stated.

Delta CO_2e = (Annual Fuel Spend ÷ Avg Price) × ICE Factor – (Mileage × EV Efficiency × Grid Factor)

Unique Differentiators

- **100% auditable AI:** every recommendation carries its XGBoost propensity score, SHAP reason codes and GHG-compliant calculation.
- **Decoupled intelligence:** the LLM has no authority over eligibility or maths, so there is no hallucination-driven compliance liability for the bank.
- **Proactive hyper-personalisation:** clear, data-backed reason codes instead of generic marketing.
- **Memory-aware advisory (optional FR9):** a vector store remembers dismissals and adapts (e.g. pivots to home insulation if solar was rejected) without overriding ML, rules or carbon outputs.

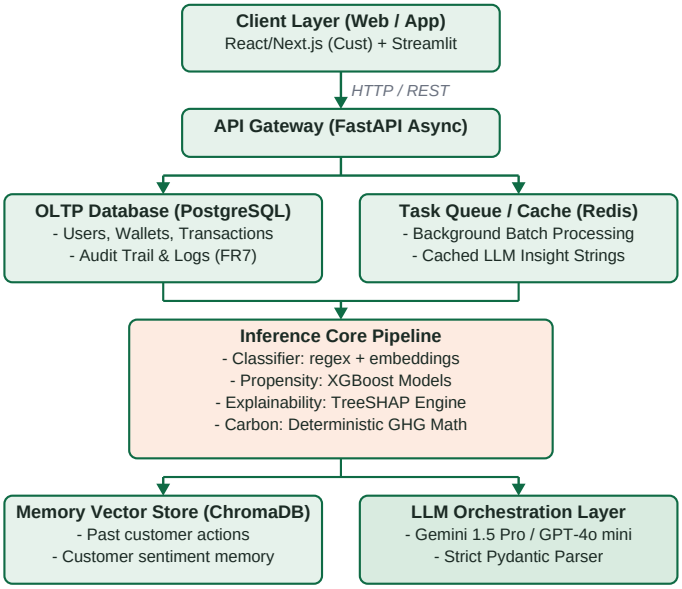
3. HIGH-LEVEL APPROACH, PIPELINE AND ARCHITECTURE

Decision pipeline

1. Categorise (FR2) Mutually exclusive tags (Fuel, Electricity, Gas, Public Transport, Home Improvement, Automotive, Utilities, Other); boundary rules documented.	**2. Feature store (FR1)** One Customer 360 view: rolling totals, utility trends, existing products.	**3. Propensity (FR3)** XGBoost gives a 0.0–1.0 score and confidence; TreeSHAP gives reason codes.
4. Rules gatekeeper Product-specific eligibility checks drop invalid offers (e.g. EV owner).	**5. Carbon engine (FR4)** Exact CO_2e reduction for surviving opportunities only.	**6. LLM + UI (FR5–FR7)** Structured data becomes plain-language insight on the Customer App and Business Dashboard.

Scalable system architecture

An asynchronous, layered architecture keeps heavy data processing (batch inference, LLM calls) away from the API routing layer, so the platform stays responsive across thousands of concurrent transactions.



Tech stack justification

- **FastAPI:** fast ASGI async framework with native Pydantic validation and auto-generated Swagger UI.
- **PostgreSQL:** ACID compliance for financial data and balances; JSONB for dynamic feature sets.
- **Redis:** queue for background batch scoring and cache for LLM insight strings to cut cost and latency.
- **XGBoost / LightGBM:** best-in-class on tabular data and supports TreeSHAP for sound reason codes.
- **ChromaDB / FAISS:** lightweight local vector store for customer memory, with no external API dependency.
- **LLM layer:** Gemini / GPT-4o mini behind a strict Pydantic parser, temperature 0.
- **Streamlit + React/Next.js:** Streamlit for the Plotly business dashboard; React for the customer app.

4. PLANNED TESTING METHODOLOGY

- **Data and splits:** synthetic data with documented schema, generator and label rules (labels depend plausibly on features, never arbitrary); customer-level 70/15/15 train/validation/test split, with all feature aggregation fitted on train only to prevent leakage.
- **ML quality vs baseline:** XGBoost compared with a simple baseline (rule thresholds / logistic regression) on held-out data using ROC-AUC, PR-AUC, precision@k and calibration (Brier score); transaction classifier scored by accuracy and macro-F1; SHAP reason codes sanity-checked per product.
- **Carbon engine:** unit tests against hand-calculated cases, with factors, units and sources versioned alongside the code.
- **LLM evaluation set:** fixed prompt set measuring JSON-schema validity, **exact numeric match** with the engine, absence of invented facts, and fallback-trigger rate.
- **Scenario tests:** missing or conflicting data; products that must not be recommended (e.g. EV owner, renter for Solar); insufficient impact data; changed customer circumstances; dismissal memory not overriding ML, rules or carbon outputs.
- **Reproducibility and responsible AI:** fixed seeds, run instructions and a Responsible AI note covering privacy, bias, explainability, hallucination, human oversight and data limitations.